INDEPENDENCE OF THE CHERN CLASS, CHERN CHARACTER AND CHERN POLYNOMIALS OF A BOUNDED FINITE LOCALLY FREE COMPLEX REPRESENTING A PERFECT OBJECT

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file chow.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with \input{preamble} and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-chow/preamble.tex is excerpted verbatim as two spans, lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. Same-chapter pointers are located at their \label lines inside the pinned chapter file; cross-chapter pointers are located in the project's own tag list staged in the snapshot (sources/stacks-chow/tags.txt), at the verbatim lines carrying their canonical tags.

Allowed standard prerequisites (editorial).

- Additivity of the total Chern class and of the Chern character. Same chapter chow.tex, labels lemma-additivity-chern-classes (used at proof lines 9111 and 9167) and lemma-chern-character-additive (used at proof line 9109). Let the product formula for the total Chern class of a complex be reorganized as a product over kernels and images, and the same for the Chern character; the reductions themselves are inside the counted span.
- Extension of the construction from finite locally free sheaves to complexes. Same chapter chow.tex, label remark-extend-to-finite-locally-free, invoked at proof line 9114. Fixes what c of a locally bounded complex of finite locally free modules is supposed to mean, so that independence of the representing complex is a well posed question.
- Splitting of the bivariant theory over an open and closed decomposition, and vanishing of a bivariant class. Same chapter chow.tex, labels lemma-disjoint-decomposition-bivariant (proof line 9131) and lemma-bivariant-zero (proof line 9141). Gives A^p(X) as the product of the A^p(X_i) so that the components can be glued, and reduces equality of bivariant classes to equality after capping with [Y] for every integral Y.
- Tor dimension at most one after a proper birational modification, and Schanuel's lemma. More on Flatness, labels flat-lemma-blowup-complex-integral (proof line 9148) and flat-remark-when-you-have-a-complex (proof line 9163); Algebra, label algebra-lemma-Schanuel (proof line 9203). Reduces to complexes of constant rank whose cohomology has tor dimension

Date: stacks-project:chow.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

at most one, in which case kernels and images are finite locally free, and identifies the two kernels in the final four-term diagram chase.

Source reference. Original lemma 00O3: sources/stacks-chow/tags.txt, source lines 881–881. Navigation pointer for \ref{algebra-lemma-Schanuel} used inside the counted span (line 9203); the argument that consumes it is retained in the span.

Source reference. Original lemma 0ESR: sources/stacks-chow/tags.txt, source lines 18491–18491. Navigation pointer for \ref{flat-lemma-blowup-complex-integral} used inside the counted span (line 9148); the argument that consumes it is retained in the span.

Source reference. Original lemma 0F8J: sources/stacks-chow/tags.txt, source lines 19044–19044. Navigation pointer for \ref{flat-remark-when-you-have-a-complex} used inside the counted span (line 9163); the argument that consumes it is retained in the span.

Source reference. Original lemma 02UI: sources/stacks-chow/pinned-original-source.tex, source lines 7965–7965. Navigation pointer for \ref{lemma-additivity-chern-classes} used inside the counted span (line 9111); the argument that consumes it is retained in the span.

Source reference. Original lemma 02UC: sources/stacks-chow/pinned-original-source.tex, source lines 6585–6585. Navigation pointer for \ref{lemma-bivariant-zero} used inside the counted span (line 9141); the argument that consumes it is retained in the span.

Source reference. Original lemma 0F9C: sources/stacks-chow/pinned-original-source.tex, source lines 8958–8958. Navigation pointer for \ref{lemma-chern-character-additive} used inside the counted span (line 9109); the argument that consumes it is retained in the span.

Source reference. Original lemma 0FDZ: sources/stacks-chow/pinned-original-source.tex, source lines 6623–6623. Navigation pointer for \ref{lemma-disjoint-decomposition-bivariant} used inside the counted span (line 9131); the argument that consumes it is retained in the span.

Source reference. Original lemma 0ESW: sources/stacks-chow/pinned-original-source.tex, source lines 7632–7632. Navigation pointer for \ref{remark-extend-to-finite-locally-free} used inside the counted span (line 9114); the argument that consumes it is retained in the span.

Source reference. Original lemma 02QL: sources/stacks-chow/pinned-original-source.tex, source lines 1372–1372. Navigation pointer for \ref{situation-setup} used inside the counted span (line 9092); the argument that consumes it is retained in the span.

Lemma 0.1. *Let (S, δ) be as in Situation 02QL. Let X be locally of finite type over S. Let $E \in D(\mathcal{O}_X)$ be an object such that there exists a locally bounded complex $\mathcal{E}^\bullet$ of finite locally free $\mathcal{O}_X$-modules representing E. Then a slight generalization of the above constructions*

$$c(\mathcal{E}^\bullet) \in \prod\nolimits_{p \geq 0} A^p(X), \quad ch(\mathcal{E}^\bullet) \in \prod\nolimits_{p \geq 0} A^p(X) \otimes \mathbf{Q}, \quad P_p(\mathcal{E}^\bullet) \in A^p(X)$$

are independent of the choice of the complex $\mathcal{E}^\bullet$.

Proof. We prove this for the total Chern class; the other two cases follow by the same arguments using Lemma 0F9C instead of Lemma 02UI.

As in Remark 0ESW in order to define the total chern class $c(\mathcal{E}^\bullet)$ we decompose X into open and closed subschemes

$$X = \coprod\nolimits_{i \in I} X_i$$

such that the rank $\mathcal{E}^n$ is constant on X_i for all n and i. (Since these ranks are locally constant functions on X we can do this.) Since $\mathcal{E}^\bullet$ is locally bounded, we see that only a finite number of the sheaves $\mathcal{E}^n|_{X_i}$ are nonzero for a fixed i. Hence we can define

$$c(\mathcal{E}^\bullet|_{X_i}) = \prod\nolimits_n c(\mathcal{E}^n|_{X_i})^{(-1)^n} \in \prod\nolimits_{p \geq 0} A^p(X_i)$$

as above. By Lemma 0FDZ we have $A^p(X) = \prod_i A^p(X_i)$. Hence for each $p \in \mathbf{Z}$ we have a unique element $c_p(\mathcal{E}^\bullet) \in A^p(X)$ restricting to $c_p(\mathcal{E}^\bullet|_{X_i})$ on X_i for all i.

Suppose we have a second locally bounded complex $\mathcal{F}^\bullet$ of finite locally free $\mathcal{O}_X$-modules representing E. Let $g : Y \to X$ be a morphism locally of finite type with Y integral. By Lemma 02UC it suffices to show that with $c(g^*\mathcal{E}^\bullet) \cap [Y]$ is the same as $c(g^*\mathcal{F}^\bullet) \cap [Y]$ and it even suffices to prove this after replacing Y by an integral scheme proper and birational over Y. Then first we conclude that $g^*\mathcal{E}^\bullet$ and $g^*\mathcal{F}^\bullet$ are bounded complexes of finite locally free $\mathcal{O}_Y$-modules of constant rank. Next, by More on Flatness, Lemma 0ESR we may assume that $H^i(Lg^*E)$ is perfect of tor dimension ≤ 1 for all $i \in \mathbf{Z}$. This reduces us to the case discussed in the next paragraph.

Assume X is integral, $\mathcal{E}^\bullet$ and $\mathcal{F}^\bullet$ are bounded complexes of finite locally free modules of constant rank, and $H^i(E)$ is a perfect $\mathcal{O}_X$-module of tor dimension ≤ 1 for all $i \in \mathbf{Z}$. We have to show that $c(\mathcal{E}^\bullet) \cap [X]$ is the same as $c(\mathcal{F}^\bullet) \cap [X]$. Denote $d^i_\mathcal{E} : \mathcal{E}^i \to \mathcal{E}^{i+1}$ and $d^i_\mathcal{F} : \mathcal{F}^i \to \mathcal{F}^{i+1}$ the differentials of our complexes. By More on Flatness, Remark 0F8J we know that $\mathrm{Im}(d^i_\mathcal{E})$, $\mathrm{Ker}(d^i_\mathcal{E})$, $\mathrm{Im}(d^i_\mathcal{F})$, and $\mathrm{Ker}(d^i_\mathcal{F})$ are finite locally free $\mathcal{O}_X$-modules for all i. By additivity (Lemma 02UI) we see that

$$c(\mathcal{E}^\bullet) = \prod\nolimits_i c(\mathrm{Ker}(d^i_\mathcal{E}))^{(-1)^i} c(\mathrm{Im}(d^i_\mathcal{E}))^{(-1)^i}$$

and similarly for $\mathcal{F}^\bullet$. Since we have the short exact sequences

$$0 \to \mathrm{Im}(d^i_\mathcal{E}) \to \mathrm{Ker}(d^i_\mathcal{E}) \to H^i(E) \to 0 \quad \text{and} \quad 0 \to \mathrm{Im}(d^i_\mathcal{F}) \to \mathrm{Ker}(d^i_\mathcal{F}) \to H^i(E) \to 0$$

we reduce to the problem stated and solved in the next paragraph.

Assume X is integral and we have two short exact sequences

$$0 \to \mathcal{E}' \to \mathcal{E} \to \mathcal{Q} \to 0 \quad \text{and} \quad 0 \to \mathcal{F}' \to \mathcal{F} \to \mathcal{Q} \to 0$$

with $\mathcal{E}$, $\mathcal{E}'$, $\mathcal{F}$, $\mathcal{F}'$ finite locally free. Problem: show that $c(\mathcal{E})c(\mathcal{E}')^{-1} \cap [X] = c(\mathcal{F})c(\mathcal{F}')^{-1} \cap [X]$. To do this, consider the short exact sequence

$$0 \to \mathcal{G} \to \mathcal{E} \oplus \mathcal{F} \to \mathcal{Q} \to 0$$

defining $\mathcal{G}$. Since $\mathcal{Q}$ has tor dimension ≤ 1 we see that $\mathcal{G}$ is finite locally free. A diagram chase shows that the kernel of the surjection $\mathcal{G} \to \mathcal{F}$ maps isomorphically to $\mathcal{E}'$ in $\mathcal{E}$ and the kernel of the surjection $\mathcal{G} \to \mathcal{E}$ maps isomorphically to $\mathcal{F}'$ in $\mathcal{F}$. (Working affine locally this follows from or is equivalent to Schanuel's lemma, see Algebra, Lemma 00O3.) We conclude that

$$c(\mathcal{E})c(\mathcal{F}') = c(\mathcal{G}) = c(\mathcal{F})c(\mathcal{E}')$$

as desired. □